

PATNA, India

WMO: 424920

Lat: 25.591N

Lon: 85.088E

Elev: 170

StdP: 14.61

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.9	48.0	39.1	35.2	79.1	41.7	39.2	68.8	12.7	68.6	11.2	68.1	0.9	270	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.7	106.4	73.7	103.9	74.2	100.7	75.5	84.4	93.5	83.6	92.2	83.0	91.0	7.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.5	170.3	88.2	81.7	165.9	87.5	81.0	161.8	86.8	48.9	93.7	48.1	92.6	47.4	91.3	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.8	11.5	41.6	110.6	2.7	2.5	39.7	112.4	38.1	113.9	36.6	115.3	34.6	117.2		
			41.2	86.1	2.5	1.0	39.4	86.9	38.0	87.4	36.6	88.0	34.8	88.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.8	60.0	67.8	77.6	86.5	89.3	89.5	86.2	86.0	85.0	81.2	72.7	63.4
	DBStd	10.70	4.69	4.45	4.94	3.92	4.32	4.43	2.90	2.39	2.80	3.48	3.84	4.33
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	260	161	19	0	0	0	0	0	0	0	0	0	80
	CDD50	10513	313	498	855	1095	1217	1186	1121	1116	1050	966	682	414
	CDD65	5298	8	97	391	645	752	736	656	651	600	501	232	29
	CDH74	67486	68	823	4237	8903	11161	10846	8637	8492	7423	5040	1630	226
	CDH80	35365	1	184	2120	5347	6943	6593	4249	4078	3325	2051	463	11
Wind	WSAvg	3.9	2.4	3.2	3.9	5.4	6.3	6.0	5.5	4.9	4.4	2.2	1.4	1.5
Precipitation	PrecAvg	46.0	0.5	0.5	0.5	0.4	1.7	4.8	14.4	10.7	8.6	3.5	0.3	0.2
	PrecMax	74.3	1.2	2.7	2.2	1.6	5.2	11.9	31.5	18.7	19.8	14.7	2.8	1.3
	PrecMin	22.8	0.0	0.0	0.0	0.0	0.0	0.1	2.5	2.8	1.0	0.0	0.0	0.0
	PrecStd	12.8	0.4	0.6	0.7	0.5	1.3	3.2	7.0	4.6	5.0	3.9	0.7	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	88.1	100.4	107.9	109.6	109.6	98.6	96.5	95.4	94.9	89.3	81.0
		MCWB	63.7	68.6	69.2	70.5	71.6	74.5	82.5	82.8	82.4	79.9	73.5	66.3
	2%	DB	75.6	84.6	96.4	104.4	107.2	105.9	95.9	94.7	93.9	92.5	86.6	78.9
		MCWB	61.8	66.9	69.5	70.3	72.7	78.1	82.1	82.6	82.0	79.0	71.5	64.3
	5%	DB	73.5	82.2	93.3	102.2	104.1	102.3	94.1	93.2	92.8	91.0	85.0	76.7
		MCWB	60.7	65.9	69.4	70.9	74.3	79.7	82.1	82.4	81.7	78.1	70.8	63.1
	10%	DB	71.2	79.1	91.0	99.0	100.6	98.8	92.4	91.5	91.2	89.3	82.7	73.8
		MCWB	59.9	64.2	69.0	71.4	76.6	80.5	82.1	82.0	81.3	77.0	69.5	62.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.7	72.0	76.4	81.2	83.9	85.0	84.8	85.1	84.7	83.4	77.7	69.6
		MCDB	74.1	82.8	89.6	94.4	97.6	96.2	92.8	92.2	92.4	90.3	84.2	77.1
	2%	WB	64.6	69.5	74.1	79.5	82.5	84.1	83.8	84.1	83.6	81.4	74.7	67.2
		MCDB	71.7	80.9	87.4	92.8	96.2	96.2	91.5	91.3	90.8	88.2	82.3	74.9
	5%	WB	63.0	67.6	72.3	78.0	81.4	83.2	83.2	83.3	82.7	80.2	72.9	65.4
		MCDB	70.0	78.2	86.3	91.1	94.4	94.2	90.5	89.9	89.3	86.8	80.9	72.8
	10%	WB	61.2	66.1	70.5	76.6	80.3	82.5	82.6	82.7	82.0	79.2	71.6	63.8
		MCDB	68.1	76.2	85.5	89.5	92.5	92.3	89.4	88.7	88.1	85.5	79.7	70.8
Mean Daily Temperature Range	5% DB	MDBR	20.0	22.6	25.0	24.5	20.5	15.7	10.6	10.5	11.1	16.0	21.7	21.6
		MCDBR	24.2	24.9	27.2	29.0	26.0	21.6	13.5	12.4	13.4	16.7	21.6	24.6
		MCWBR	11.6	10.0	8.8	9.5	8.2	6.5	4.4	4.2	4.8	5.9	8.6	11.4
	5% WB	MCDBR	20.3	22.5	23.9	21.7	19.6	16.6	11.9	11.3	12.0	14.2	19.2	20.8
		MCWBR	11.0	10.4	9.3	9.0	7.8	6.2	4.9	4.7	5.1	5.9	8.8	10.9
Clear-Sky Solar Irradiance	taub	0.597	0.548	0.545	0.622	0.708	0.723	0.622	0.589	0.589	0.622	0.690	0.676	
	taud	1.658	1.810	1.850	1.740	1.616	1.620	1.839	1.940	1.914	1.768	1.572	1.560	
	Ebn at Noon	203	232	242	227	208	204	225	232	228	209	178	175	
	Edh at Noon	70	64	65	74	84	83	67	60	60	66	76	75	
All-Sky Solar Radiation	RadAvg	991	1414	1820	2051	2025	1731	1425	1492	1460	1433	1204	960	
	RadStd	116	96	61	94	103	117	125	74	104	101	62	112	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page